

Requirements Engineering Report

AIU Moodle Learning Management System

Student Name

Student ID

Lab Group

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1 Introduction

Requirements Engineering (RE) is a systematic process used to identify, analyse, document, validate, and manage software system requirements. It ensures that software systems satisfy stakeholder expectations and operational objectives while maintaining quality, reliability, and usability.

The AIU Moodle Learning Management System (LMS) is an online educational platform used by Albukhary International University (AIU) to support teaching and learning activities. The system enables lecturers to distribute learning materials, create assessments, monitor student progress, and communicate with students. Students use the system to access lecture materials, participate in quizzes, submit assignments, receive announcements, and monitor academic performance.

The increasing reliance on digital learning platforms makes requirements engineering highly important in ensuring that LMS systems remain efficient, reliable, secure, and user-friendly. Therefore, this report analyses the AIU Moodle LMS using requirements engineering techniques and tools to produce a structured software requirements analysis.

1.1 Objectives of the Report

The objectives of this report are:

1. To analyse the software requirements of the AIU Moodle LMS.
2. To identify and document functional and non-functional requirements.
3. To apply requirements engineering techniques and tools for system modelling and documentation.

2 Stakeholder Analysis

Stakeholder analysis is an important process in requirements engineering because it identifies individuals and groups who interact with the system and influence system requirements.

Stakeholder	Needs and Interests
Student	Students require access to lecture notes, quizzes, assignment submissions, announcements, grades, discussion forums, and messaging functionality. Students also expect responsive mobile accessibility and reliable system availability.
Lecturer	Lecturers require functionalities for uploading learning materials, creating quizzes, grading assignments, posting announcements, and managing course activities efficiently.
School Executive	School executives are responsible for assigning students into appropriate academic groups and managing certain administrative configurations within the LMS.

3 Requirements Elicitation and Documentation

Requirements elicitation involves collecting information from stakeholders to identify system functionalities, operational expectations, and quality requirements.

Several elicitation techniques were conceptually applied during this analysis including:

1. Observation of system usage.
2. Analysis of existing Moodle functionalities.
3. User experience evaluation.
4. Stakeholder requirement identification.

3.1 Functional Requirements

Functional requirements describe the major operations and services provided by the system.

ID	Requirement	Description
FR-01	User Authentication	The system shall allow users to log in using institutional credentials or Google authentication.
FR-02	Assignment Submission	The system shall allow students to upload assignments in multiple file formats including PDF, DOCX, and ZIP files.
FR-03	Assignment Resubmission	The system shall allow students to resubmit assignments before the submission deadline.
FR-04	Quiz Participation	The system shall allow students to participate in timed online quizzes.
FR-05	Grade Viewing	The system shall allow students to view released grades and lecturer feedback.
FR-06	Material Upload	The system shall allow lecturers to upload lecture notes, rubrics, and instructional materials.
FR-07	Announcement Management	The system shall allow lecturers to publish announcements for students.
FR-08	Forum Participation	The system shall allow students and lecturers to participate in academic discussion forums.
FR-09	Course Management	The system shall allow lecturers to manage course content and assessments.
FR-10	Group Assignment Management	The system shall allow school executives to assign students into appropriate groups.

3.2 Non-Functional Requirements

Non-functional requirements describe quality attributes and operational constraints.

ID	Requirement	Description
NFR-01	Performance	The system shall maintain stable performance during peak examination and assignment submission periods.
NFR-02	Usability	The system shall provide a user-friendly interface accessible through desktop and mobile devices.

NFR-03	Security	The system shall protect academic data using secure authentication and authorization mechanisms.
NFR-04	Availability	The system shall remain accessible 24 hours a day except during scheduled maintenance.
NFR-05	Scalability	The system shall support increasing numbers of concurrent users without major performance degradation.
NFR-06	Reliability	The system shall minimize failures and maintain data consistency during user operations.

4 Use Case Modelling

Use case modelling visually represents interactions between system users and the AIU Moodle LMS.

4.1 Use Case Diagram



Figure 1: AIU Moodle LMS Use Case Diagram

4.2 Use Case Descriptions

4.2.1 UC-01 Login

Actor: Student, Lecturer, School Executive

Precondition: User possesses valid credentials.

Main Flow:

1. User accesses LMS login page.
2. User enters login credentials.
3. System validates credentials.
4. System grants access to dashboard.

Postcondition: User successfully accesses the system.

4.2.2 UC-02 Submit Assignment

Actor: Student

Precondition: Assignment submission is available.

Main Flow:

1. Student opens course page.
2. Student selects assignment.
3. Student uploads assignment file.
4. System validates submission.
5. System confirms successful upload.

Postcondition: Assignment is stored successfully.

4.2.3 UC-03 Upload Learning Materials

Actor: Lecturer

Precondition: Lecturer has course access.

Main Flow:

1. Lecturer selects course.
2. Lecturer uploads learning materials.
3. System stores uploaded materials.
4. Students gain access to materials.

Postcondition: Learning materials become accessible.

4.2.4 UC-04 Participate in Quiz

Actor: Student

Precondition: Quiz is available.

Main Flow:

1. Student opens quiz.
2. Student answers questions.
3. System records responses.
4. Quiz automatically submits when time expires.

Postcondition: Quiz responses are stored.

4.2.5 UC-05 Post Announcement

Actor: Lecturer

Precondition: Lecturer is assigned to course.

Main Flow:

1. Lecturer creates announcement.
2. Lecturer publishes announcement.
3. Students receive notification.

Postcondition: Announcement becomes visible to students.

5 Requirements Prioritisation Using MoSCoW

The MoSCoW prioritisation technique categorizes requirements according to implementation priority.

Priority	Requirement	Justification
Must Have	User Authentication	Essential for secure system access and account protection.
Must Have	Assignment Submission	Core academic functionality for coursework submission.
Must Have	Quiz Participation	Essential for online assessments and evaluations.
Must Have	Grade Viewing	Required for academic monitoring and feedback access.
Should Have	Mobile Accessibility	Improves accessibility and convenience for users.
Should Have	Forum Discussion	Enhances communication and collaborative learning.
Could Have	Enhanced Notification System	Improves announcement visibility and user awareness.
Could Have	Advanced Analytics Dashboard	Provides improved monitoring and reporting features.
Won't Have	AI-Based Recommendation System	Excluded due to current project scope limitations.

5.1 Requirements Traceability Matrix (RTM)

Requirement ID	Requirement Description	Source/Stakeholder	Related Requirement	Status
FR-01	User authentication using institutional or Google login credentials	Student, Lecturer	NFR-03 Security	Approved
FR-02	Assignment submission using multiple file formats	Student	FR-03 Assignment Resubmission	Approved
FR-03	Assignment resubmission before deadline	Student	FR-02 Assignment Submission	Approved
FR-04	Timed online quiz participation	Student, Lecturer	NFR-01 Performance	Approved
FR-05	Viewing grades and lecturer feedback	Student	FR-04 Quiz Participation	Approved
FR-06	Upload lecture materials and rubrics	Lecturer	FR-09 Course Management	Approved
FR-07	Publish course announcements	Lecturer	NFR-02 Usability	Proposed
FR-08	Participate in forum discussions	Student, Lecturer	NFR-02 Usability	Approved
FR-09	Manage course content and assessments	Lecturer	FR-06 Material Upload	Approved
FR-10	Assign students into academic groups	School Executive	FR-01 User Authentication	Approved
NFR-01	Maintain stable system performance during peak usage	Student, Lecturer	FR-04 Quiz Participation	Changed
NFR-02	Provide mobile-friendly and user-friendly interface	Student	FR-07 Announcement Management	Approved
NFR-03	Secure authentication and data protection	All Stakeholders	FR-01 FR-01 User Authentication	Approved
NFR-04	Ensure 24/7 system availability	All Stakeholders	NFR-01 Performance	Proposed
NFR-05	Support scalability during concurrent usage	All Stakeholders	NFR-01 Performance	Proposed
NFR-06	Ensure reliable and consistent system operation	All Stakeholders	NFR-04 Availability	Approved

5.2 Change Management and Version Control

Change management ensures that modifications to system requirements are properly documented, reviewed, approved, and tracked throughout the project lifecycle.

For the AIU Moodle LMS requirements analysis, changes were managed using version control and documentation tracking techniques. GitHub was used as the primary version control platform to record updates, maintain documentation history, and monitor requirement modifications.

Whenever a requirement was updated, the following process was followed:

1. Requirement change identified.
2. Change impact analysed.
3. Requirement documentation updated.
4. Version history recorded using GitHub commits.
5. Updated requirement status reflected in the RTM.

The version control process improved documentation consistency, prevented loss of previous requirement versions, and supported better requirement traceability.

Table 6: Requirement Change Log

Version	Date	Changes Made	Updated By
v1.0	Initial Draft	Created initial requirements documentation	Win Yan Naing Htut
v1.1	Revision 1	Added non-functional requirements and RTM	Win Yan Naing Htut
v1.2	Revision 2	Updated use case descriptions and prioritisation	Win Yan Naing Htut
v1.3	Final Version	Final formatting and documentation improvements	Win Yan Naing Htut

6 Tool Usage Explanation

Several requirements engineering tools were used to support modelling, documentation, visualization, and project organization.

6.1 Draw.io

Draw.io was selected for creating UML use case diagrams because it provides simple diagram creation tools and easy export functionality.

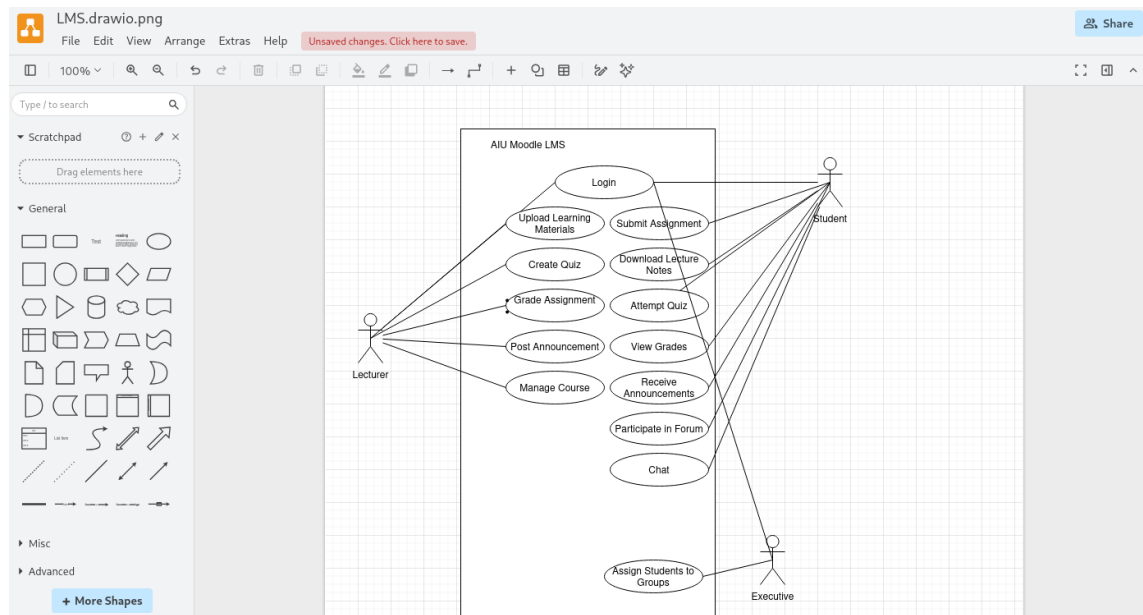


Figure 2: Draw.io

6.2 Lucidchart

Lucidchart was used to create workflow diagrams and stakeholder interaction diagrams because it supports collaborative visual modelling.

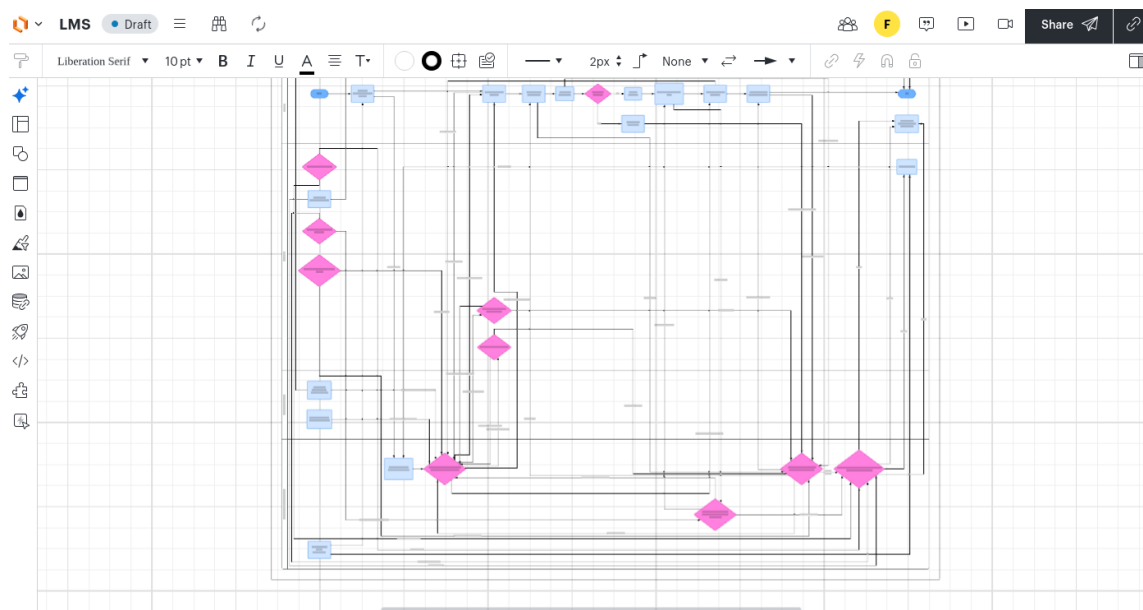


Figure 3: Lucidchart

6.3 GitHub

GitHub was used for version control and documentation management. It helped organize files and maintain report revisions systematically.

Insert GitHub screenshot here.

6.4 Trello

Trello was used to manage assignment tasks and monitor progress using a Kanban-style board.

Insert Trello screenshot here.

6.5 ReqView

ReqView was used to organize software requirements systematically and improve requirements traceability.

Insert ReqView screenshot here.

7 Conclusion

The requirements engineering analysis conducted for the AIU Moodle Learning Management System identified several important functional and non-functional requirements necessary to support effective online learning operations.

Through stakeholder analysis, requirements elicitation, use case modelling, prioritisation, and traceability analysis, this report demonstrated the application of structured requirements engineering methodologies in analysing educational software systems.

The analysis also highlighted practical system concerns such as weak notification visibility, navigation difficulties, and slow performance during examination periods. These findings contributed to the development of realistic requirements focused on improving system usability, performance, scalability, and reliability.

Furthermore, the use of professional requirements engineering tools improved documentation quality, modelling accuracy, and project organization. Overall, this report demonstrates the importance of systematic software requirements engineering in developing reliable and efficient learning management systems.

8 Reflection

This assignment provided practical experience in applying requirements engineering techniques to analyse a real-world software system. One of the major challenges encountered was identifying accurate stakeholder responsibilities, especially for administrative users such as school executives.

Another challenge involved distinguishing between functional and non-functional requirements while ensuring proper requirement structure and clarity. Since the AIU Moodle LMS is an existing operational system, analysing user experiences and identifying realistic system issues required careful observation and evaluation.

Several practical system limitations were identified during the analysis including weak notification visibility, navigation complexity, and slow performance during examination periods. These observations helped strengthen the quality of the non-functional requirements.

The selected tools significantly improved the overall requirements engineering process. Draw.io simplified UML diagram creation, Lucidchart improved workflow visualization, Trello enhanced task organization, GitHub supported documentation version management, and ReqView improved requirements organization.

Overall, this assignment improved understanding of requirements engineering methodologies and demonstrated the importance of systematic analysis, stakeholder understanding, and structured documentation in software engineering projects.

9 References

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